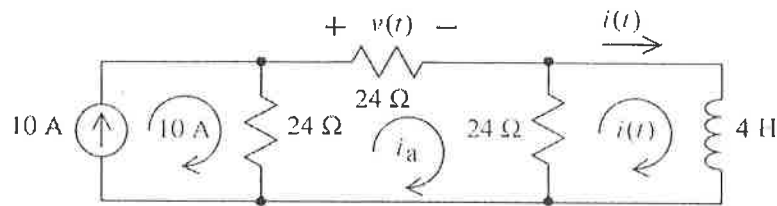


P7.5-13



15pt

We'll write and solve a mesh equation. Label the meshes as shown. Apply KVL to the center mesh to get

$$24i_a + 24(i_a - i(t)) + 24(i_a - 10) = 0 \Rightarrow i_a = \frac{i(t) + 10}{3} = 5 - e^{-4t} \text{ A for } t > 0$$

Then

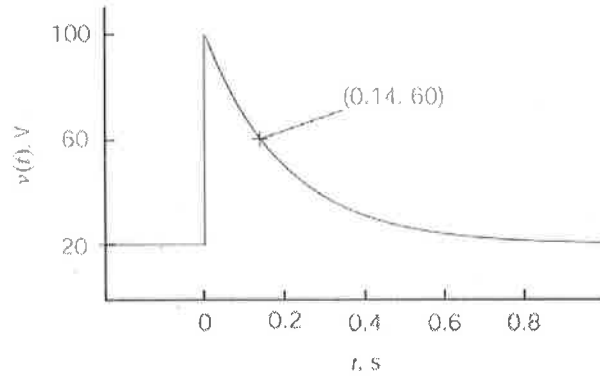
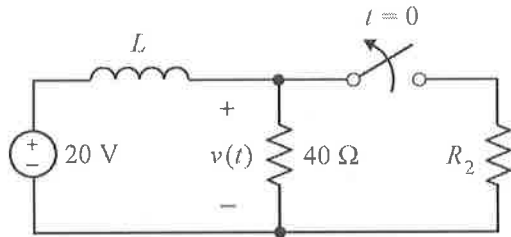
$$v(t) = 24i_a = 120 - 24e^{-4t} \text{ V for } t > 0$$

5pt.

(checked: LNAP 6/25/04)

15

P8.3-25



From the plot $D = v(t)$ for $t < 0 = 20$ V, $E + F = v(0+) = 100$ V and $E = \lim_{t \rightarrow \infty} v(t) = 20$ V. The point labeled on the plot indicates that $v(t) = 60$ V when $t = 0.14$ s. Consequently

$$60 = 20 + 80e^{-a(0.14)} \Rightarrow a = \frac{\ln\left(\frac{60-20}{80}\right)}{-0.14} = 5 \frac{1}{s}$$

Then

$$v(t) = \begin{cases} 20 \text{ V} & \text{for } t \leq 0 \\ 20 + 80e^{-5t} \text{ V} & \text{for } t \geq 0 \end{cases}$$

At $t = 0+$,

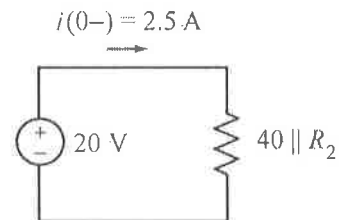
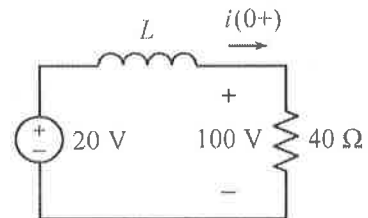
$$i(0+) = \frac{100}{40} = 2.5 \text{ A}$$

When $t < 0$, the circuit is at steady state so the inductor acts like a short circuit.

$$40 \parallel R_2 = \frac{20}{2.5} = 8 \Omega \Rightarrow R_2 = 10 \Omega$$

Next, the inductance can be determined using the time constant:

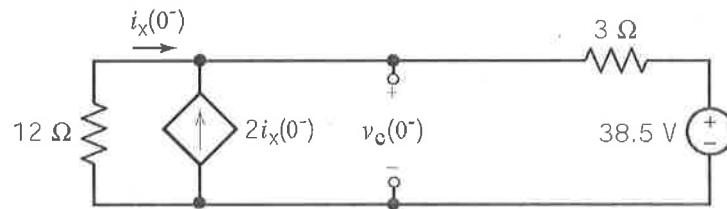
$$5 = a = \frac{1}{\tau} = \frac{40}{L} \Rightarrow L = \frac{40}{5} = 8 \text{ H}$$



20pt

P8.7-1

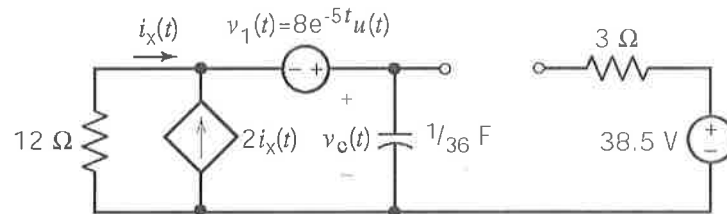
Assume that the circuit is at steady state before $t = 0$:



$$\text{KVL : } 12i_x + 3(3i_x) + 38.5 = 0 \Rightarrow i_x = -1.83 \text{ A}$$

$$\text{Then } v_c(0^-) = -12i_x = 22 \text{ V} = v_c(0^+)$$

After $t = 0$:



$$\text{KVL : } 12i_x(t) - 8e^{-5t} + v_c(t) = 0$$

$$\text{KCL : } -i_x(t) - 2i_x(t) + (1/36) \frac{dv_c(t)}{dt} = 0 \Rightarrow i_x(t) = \frac{1}{108} \frac{dv_c(t)}{dt}$$

$$\therefore 12 \left[\frac{1}{108} \frac{dv_c(t)}{dt} \right] - 8e^{-5t} + v_c(t) = 0$$

$$\frac{dv_c(t)}{dt} + 9v_c(t) = 72e^{-5t} \Rightarrow v_{cn}(t) = Ae^{-9t}$$

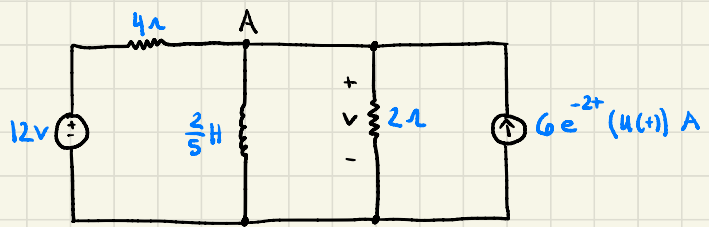
$$\text{Try } v_{cf}(t) = Be^{-5t} \text{ \& substitute into the differential equation } \Rightarrow B = 18$$

$$\therefore v_c(t) = Ae^{-9t} + 18e^{-5t}$$

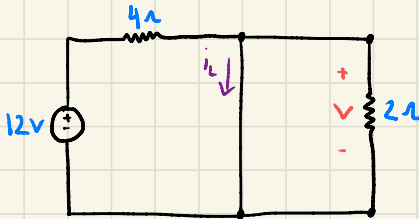
$$v_c(0) = 22 = A + 18 \Rightarrow A = 4$$

$$\therefore v_c(t) = 4e^{-9t} + 18e^{-5t} \text{ V}$$

1.)

Find $v(t)$ for $t > 0$ Before $t = 0$

- The current source is "gone".
- The inductor acts as a short.

Initial current through inductor

$$-12 + 4i = 0$$

$$i = 3A$$

After $t > 0$

Node Voltage @ A:

$$4 \cdot \left(\frac{A - 12}{4} + i_L + \frac{A}{2} - 6e^{-2t} u(t) = 0 \right) \cdot 4$$

$$A - 12 + 4i_L + 2A - 24e^{-2t} u(t) = 0$$

$$3A + 4i_L = 24e^{-2t} + 12$$

$$3 \left(\frac{2}{5} \frac{di_L}{dt} \right) + 4i_L = 24e^{-2t} + 12$$

$$\frac{di}{dt} + \frac{10}{3}i = 20e^{-2t} + 10$$

$$e^{\frac{10}{3}t} \cdot i = \int e^{\frac{10}{3}t} \cdot 20e^{-2t} + 10$$

$$i = 15e^{-2t} + 3 \cdot 15e^{-\frac{10}{3}t}$$

$$v(t) = \frac{2}{5} \left(\frac{d}{dt} 15e^{-2t} + 3 - 30e^{-\frac{10}{3}t} \right) = 20e^{-\frac{10}{3}t} - 12e^{-2t}$$

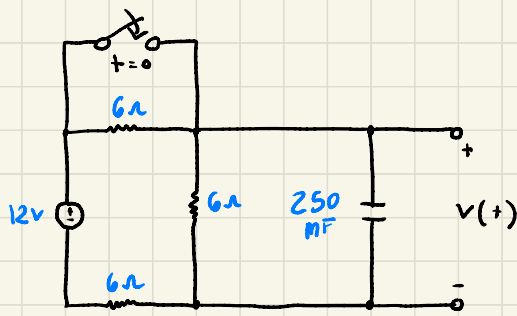
$$A = L \frac{di}{dt}$$

$$i_0 = 3A$$

$$C = -15$$

$$\int \frac{10}{3} dt = e^{\frac{10}{3}t}$$

2)



Find $v(t)$ for $t > 0$

Before $t > 0$

$V_c = V_{6\Omega}$ in Parallel

$$V_{6\Omega} = \frac{6}{18} = \frac{1}{3} \cdot 12V = 4V \quad \text{initial Voltage}$$

After $t > 0$

$V_c = V_{6\Omega}$ in Parallel

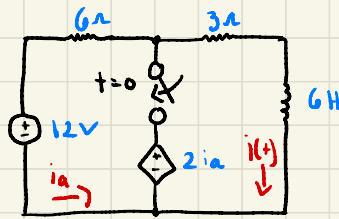
$$V_{6\Omega} = \frac{6}{12} = \frac{1}{2} \cdot 12 = 6V \quad \text{long term Voltage}$$

$$\text{Formula: } V(t) = V_{t>0} + (V_0 - V_{t>0}) e^{-t/\tau}$$

$$\tau = \frac{R}{C} = \frac{3\Omega}{250mF} = 1.33s$$

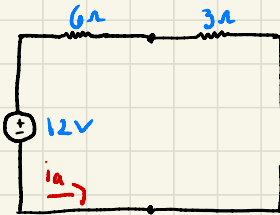
$$\begin{aligned} v(t) &= 6 + (4 - 6) e^{-1.33t} \\ &= 6 - 2e^{-1.33t} \end{aligned}$$

3.)



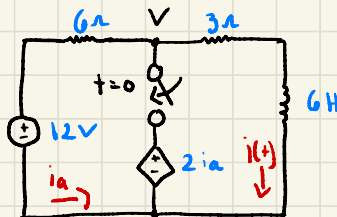
Find $i(t)$ for $t > 0$

Before $t > 0$



$$\text{Current: } \frac{12}{9} = \frac{4}{3} \text{ A}$$

After $t > 0$



$$\frac{2i_a - 12}{6} \Rightarrow i_a = -3 \text{ A}$$

Solve for τ

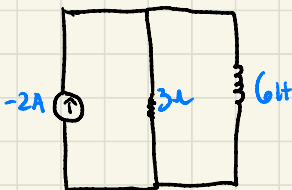
$$\tau = \frac{L}{R}$$

$$R_T = \frac{1}{6} + \frac{1}{3} = 3\Omega \quad V = 2(-3) = -6\text{V}$$

$$i_L = \frac{-6}{3} = -2 \text{ A}$$

$$\text{Formula: } i_{t>0} + (i_0 - i_{t>0})e^{-t/\tau}$$

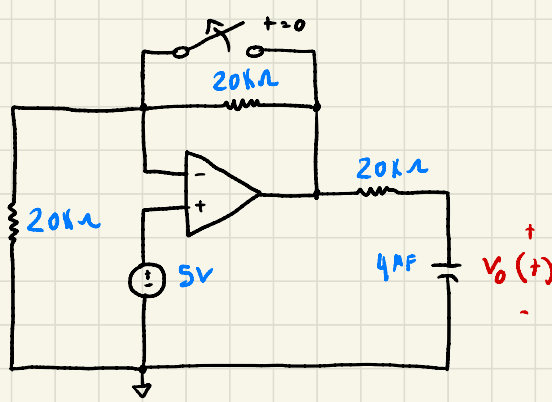
$$\frac{4}{3} + \frac{6}{3} = \frac{10}{3}$$



$$\tau = \frac{6}{3} = 2$$

$$i(t) = -2 + \left(\frac{4}{3} + 2\right)e^{-t/2} = -2 + \frac{10}{3}e^{-t/2}$$

4.)

Find $V_o(t)$ for $t > 0$

$$\text{Formula: } v(t) = V_{t=0} + (V_0 - V_{t=0}) e^{-t/\tau}$$

Before $t > 0$

$$V_0 = 5V \text{ since output resistance} = 0$$

After $t > 0$

$$i_{Rf} = \frac{5}{20k} = 0.25 \text{ mA}$$

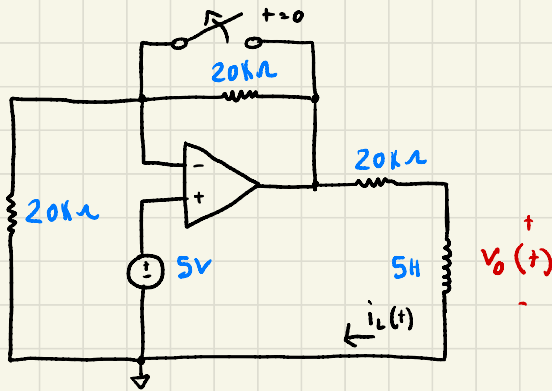
$$V_0 = V_0 - 5 = 5 = 10V$$

$$\tau = RC = 20k \cdot 4\mu F = 80 = \frac{1}{12.5} = 12.5$$

$$V_o(t) = 10 + (5 - 10) e^{-12.5t}$$

$$= 10 - 5 e^{-12.5t}$$

5)

Find $V_o(t)$ for $t > 0$ Before $t > 0$

$$V_o = 5V$$

$$i_L = \frac{5}{20k} = 0.25 \text{ mA}$$

After $t > 0$

$$V_o = 10V$$

$$i_L = \frac{10}{20k} = 0.500 \text{ mA}$$

$$\tau = \frac{L}{R} = \frac{5H}{20k} = \frac{1}{4000} = \frac{1}{4000} = 4000$$

$$i(t) = 0.500 + (0.25 - 0.500)e^{-4000t}$$

$$V_o(t) = 5 \left(\frac{d}{dt} 0.500 - 0.25e^{-4000t} \right)$$

$$= 5e^{-4000t}$$

1.(a) charge: area of just

$$A = (4 \times 5 \times 0.5 + 4 \times 10 + 8 \times 6 + 4 \times 8 \times 0.5 + 3 \times 6 \times 0.5) \times 10^3 \\ = (10 + 40 + 48 + 16 + 9) \times 10^3 = \underline{\underline{123 \times 10^3 \text{ C}}}$$

(b) Energy: $W = \int v i dt$. Get Power function.

$$v = \frac{3}{15000}t + 9 = 0.2 \times 10^{-3}t + 9 \quad (0 < t < 15 \text{ ks})$$

$$i = -1.25 \times 10^{-3}t + 15 \quad (0 < t < 4 \text{ ks})$$

$$i = -0.5 \times 10^{-3}t + 12 \quad (4 < t < 12 \text{ ks})$$

$$i = -2 \times 10^{-3}t + 30 \quad (12 < t < 15)$$

$$P \propto \underline{vi}$$

$$0 < t < 4 \text{ ks} ; P = (0.2 \times 10^{-3}t + 9) \cdot (-1.25 \times 10^{-3}t + 15) \\ = -0.25 \times 10^{-6}t^2 - 8.25 \times 10^{-3}t + 135$$

$$W = \int_0^{4000} P dt = 468.667 \text{ kJ}$$

$$4 < t < 12 ; P = -0.1 \times 10^{-6}t^2 - 2.1 \times 10^{-3}t + 108$$

$$W = \int_{4000}^{12000} P dt = 674.133 \text{ kJ}$$

$$12 < t < 15 ; P = -0.4 \times 10^{-6}t^2 - 12 \times 10^{-3}t + 270$$

$$W = \int_{12000}^{15000} P dt = 104.4 \text{ kJ}$$

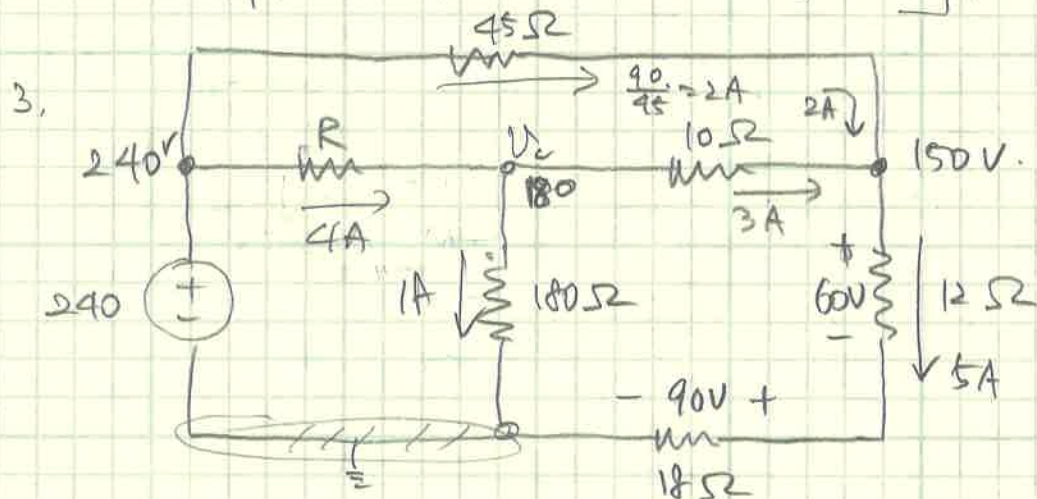
$$\therefore \text{total energy} \propto 1247.2 \text{ kJ}$$

2 (a) left loop K.V.L. $-5 + 54000 i_1 - 1 + 6000 (31 i_1) = 0$
 $-6 + 240000 i_1 = 0 \quad i_1 = 2.5 \times 10^{-5} \text{ A} = \underline{25 \mu\text{A}}$

(b) Big loop $-5 + 54000 \times i_1 - 1 + V - 1800 \times 30 \times i_1 + 8 = 0$
 $2 + 54000 \times 25 \times 10^{-6} + V - 54000 \times i_1 = 0$
 $\underline{V = -2 \text{ V}}$

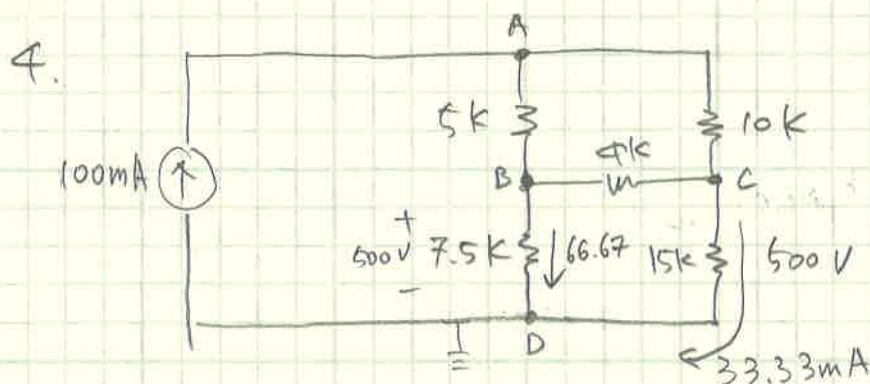
(c) 5V: $5 \times 25 \times 10^{-6}$ supply 125×10^{-6}
 1V: $1 \times 25 \times 10^{-6}$ supply 25×10^{-6}
 $30 i_1$: $2 \times 30 \times 25 \times 10^{-6}$ absorb
 8V: $8 \times 30 \times 25 \times 10^{-6} = 6000 \times 10^{-6}$ supply
 $\therefore (6000 + 125 + 25) \times 10^{-6}$ supply $\boxed{6150 \mu\text{W}}$

(d) $54000 \times (25 \times 10^{-6})^2 = 33.75 \mu\text{W}$
 $1800 \times (30 \times 25 \times 10^{-6})^2 = 1012.5 \mu\text{W}$
 $6000 \times (31 \times 25 \times 10^{-6})^2 = 3603.75 \mu\text{W}$
 $30 i_1$: $2 \times 30 \times 25 \times 10^{-6} = 1500 \mu\text{W}$ $\boxed{+ 6150 \mu\text{W}}$



$$V_c - 150 = 30 \quad V_c = 180$$

$$(240 - 180) / 4 = 60 / 4 = \underline{15 \Omega}$$



$$I_{BD} + I_{CD} = 100 \text{ mA}$$

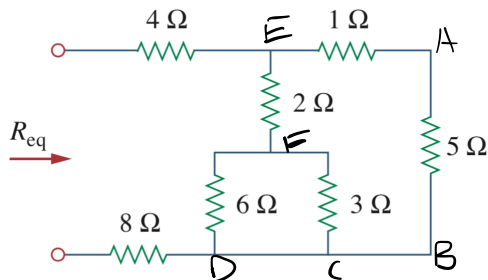
$$I_{BC} = 100 - 33.33 = 66.67 \text{ mA}$$

$$V_B = V_C \rightarrow I_{BC} = 0.$$

$$\therefore \left\{ \begin{array}{l} P_{5k} = (66.67 \text{ mA} \cdot 5k\Omega) \cdot 66.67 \text{ mA} = 22.22 \text{ W} \\ P_{10k} = (33.33 \text{ mA} \cdot 10k\Omega) \cdot 33.33 \text{ mA} = 11.10 \text{ W} \\ P_{4k} = 0. \quad P_{15k} = 500V \times 33.33 \text{ mA} = 16.67 \text{ W} \\ P_{7.5k} = 500 \times 66.67 = 33.33 \text{ W} \end{array} \right.$$

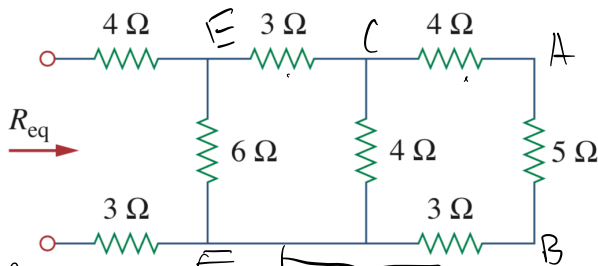
$$P_{7.5k} = 500 \times 66.67 = 33.33 \text{ W}.$$

Find equivalent resistance

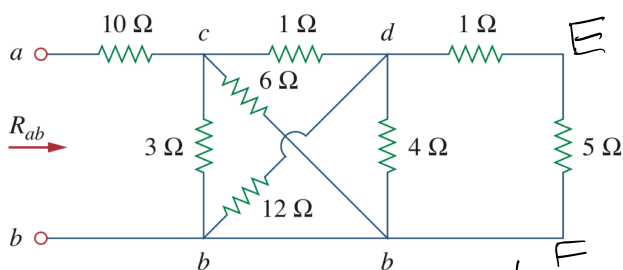


$$R_{eq} = 8 + 4 + 2 \cdot 4 = 14.4 \Omega$$

Find equivalent resistance

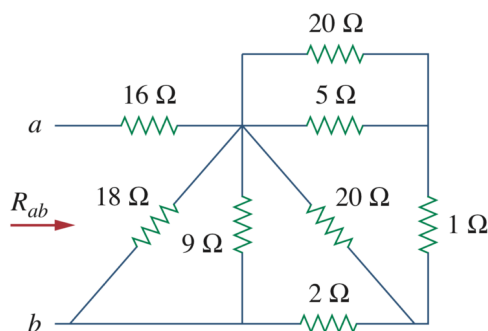


$$R_{eq} = 4 + 3 + 3 = 10 \Omega$$

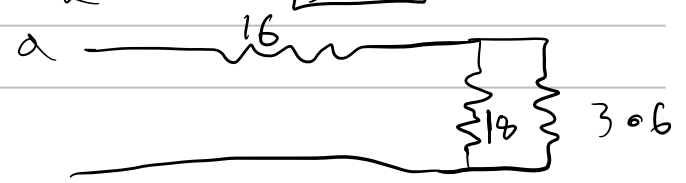
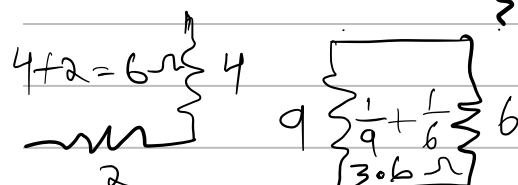
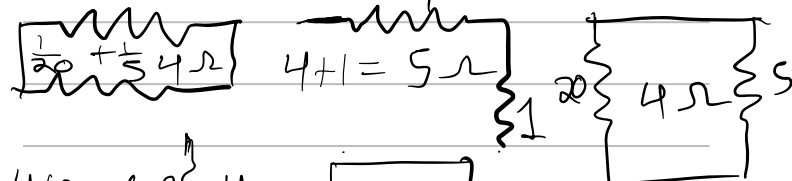
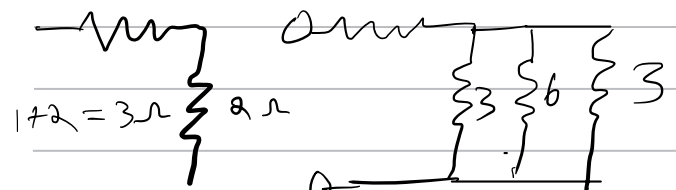
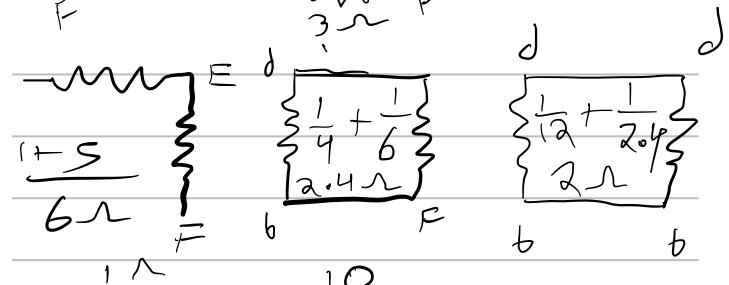
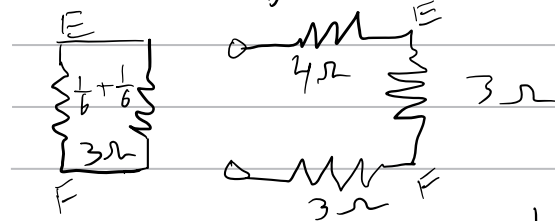
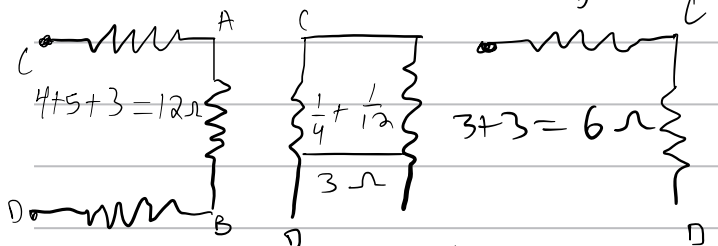
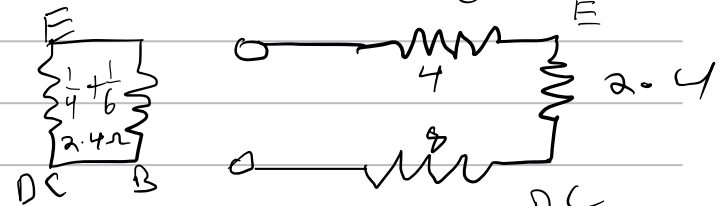
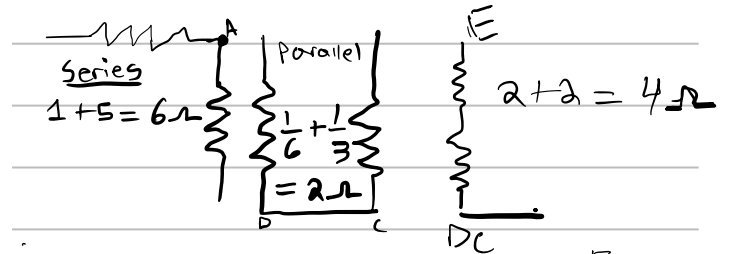


$$R_{ab} = 10 + \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{3} \right)^{-1} = 11.2 \Omega$$

Find equivalent resistance



$$R_{ab} = 16 + \left(\frac{1}{9} + \frac{1}{3 \cdot 6} \right)^{-1} = 19 \Omega$$



P 3.2-11 Determine the power received by each of the resistors in the circuit shown in Figure P 3.2-11.

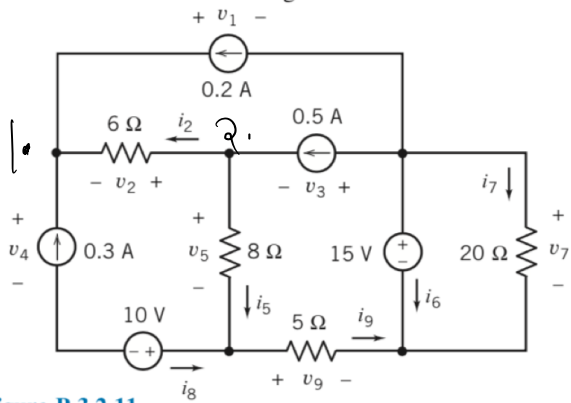


Figure P 3.2-11

Ranvir Thind
HW 3A
05-08-26

1. KCL $0.3 \text{ A} + 0.2 \text{ A} + I_2 = 0 \rightarrow I_2 = -0.5 \text{ A}$
 $P_{6\Omega} = (-0.5)^2 (6) = \boxed{1.5 \text{ W}}$

2. KCL $-0.5 \text{ A} - 0.5 \text{ A} - I_5 = 0$
 $I_5 = -1 \text{ A}$ $P_{8\Omega} = (-1)^2 (8) = \boxed{8 \text{ W}}$

3. KCL $I_8 + I_5 + I_9 = 0$ $I_8 = 0.3$
 $I_9 = -0.7$

$P_{5\Omega} = (0.7)^2 (5) = \boxed{2.45 \text{ W}}$

4. $P_{20\Omega} = \frac{V^2}{R} = \frac{15^2}{20} = \boxed{11.25 \text{ W}}$

P 3.2-11 Determine the power received by each of the resistors in the circuit shown in Figure P 3.2-11.

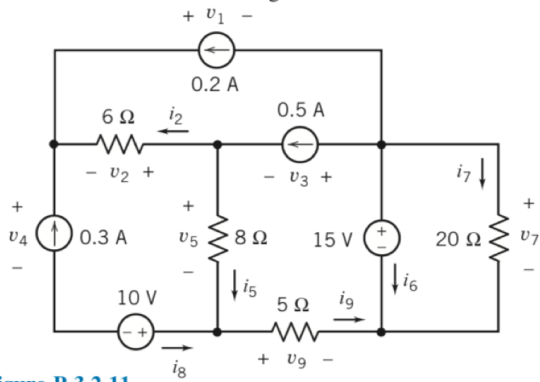


Figure P 3.2-11

$$I_1 = 0.75 \text{ A} + 0.25 \text{ A} = 1 \text{ A} \quad V_1 = 1 \text{ A} \cdot 20 \Omega = 20 \text{ V}$$

$$I_2 = 0.25 - 1 = -0.75 \text{ A} \quad V_2 = -0.75 \cdot 10 = -7.5 \text{ V}$$

$$V_3 = 15 \text{ V} \quad I_3 = \frac{15}{60} = 0.25 \text{ A}$$

$$15 \text{ V}$$

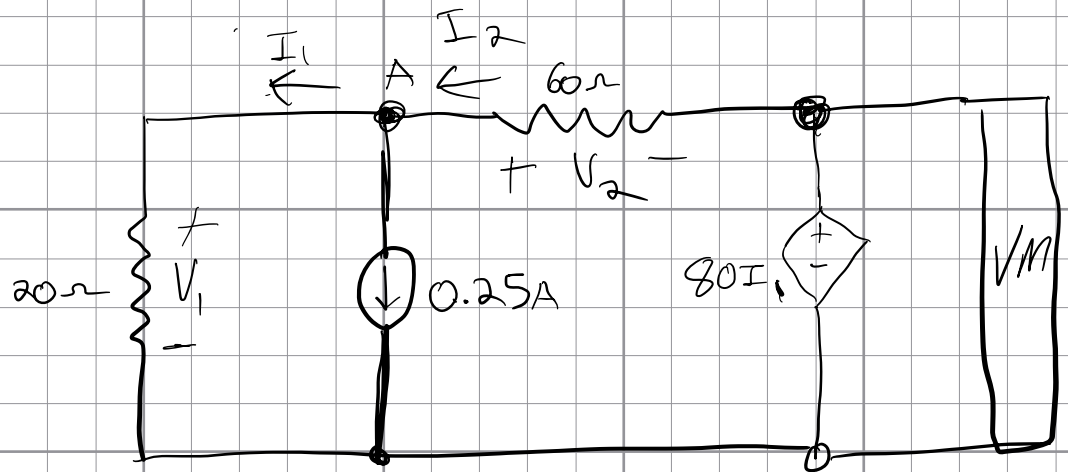
$$I_4 = 0.75 + 0.25 \text{ A} = 1 \text{ A}$$

$$0.75 \text{ A}$$



$$0.25 \text{ A}$$

$$V = 20 \text{ V} \quad V = 20 - 7.5 = 12.5 \text{ V} \rightarrow \text{parallel } 20 \Omega \text{ resistor}$$



Apply KCL at node A

$$I_2 = I_1 + 0.25$$

Find V_1 and V_2

$$V_1 = 20 I_1 \quad V_2 = 60(I_1 + 0.25)$$

Apply KVL

$$\begin{aligned} V_1 + V_2 + 80 I_1 &= 0 \\ 20 I_1 + 60 I_1 + 15 + 80 I_1 &= 0 \\ I_1 &= -0.09375 \text{ A} \end{aligned}$$

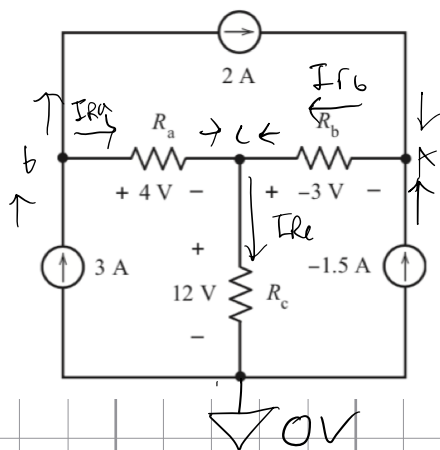
Solve for V_m

$$V_m = 80 I_1 = 80(-0.09375 \text{ A})$$

$$V_m = \boxed{7.5 \text{ V}}$$

P 3.2-27 $\oplus$ Consider the circuit shown in Figure P 3.2-27.

- Determine the values of the resistances.
- Determine the values of the power supplied by each current source.
- Determine the values of the power received by each resistor.



KCL @ Node A

(a) $I_{R_b} = 2 - 1.5 \text{ A} = 0.5 \text{ A}$

$$R_b = \left| \frac{-3 \text{ V}}{0.5 \text{ A}} \right| = 6 \Omega$$

KCL @ Node B

$$I_{R_a} = 3 - 2 = 1 \text{ A}$$

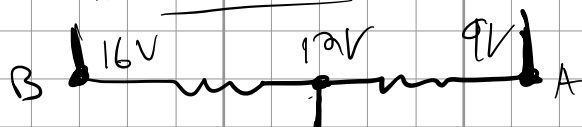
$$R_a = \left| \frac{4}{1} \right| = 4 \Omega$$

KCL @ Node C

$$I_{R_c} = 1 \text{ A} + 0.5 = 1.5 \text{ A}$$

$$R_c = \left| \frac{12}{1.5} \right| = 8 \Omega$$

(b) 2A Source



$$P = VI \quad \Delta V = 16 - 9 = 7 \text{ V}$$

$$P = 7 \times 2 = 14 \text{ W}$$

3A Source

$$\Delta V = 16 \text{ V} - 0 = 16 \text{ V}$$

$$P = 16 \times 3 = 48 \text{ W}$$

-1.5A Source

$$\Delta V = 9 - 0 = 9 \text{ V}$$

$$P = 9 \times 1.5 = 13.5 \text{ W}$$

(c) Ra

$$P = I_{R_a}^2 R_a = 0.5^2 \times 4 = 1.0 \text{ W}$$

Rb

$$P = I_{R_b}^2 R_b = 0.5^2 \times 6 = 1.5 \text{ W}$$

Rc

$$P = I_{R_c}^2 R_c = 1.5^2 \times 8 = 18 \text{ W}$$

P 3.2-28 $\oplus$ Consider the circuit shown in Figure P 3.2-28.

- Determine the value of the power supplied by each independent source.
- Determine the value of the power received by each resistor.
- Is power conserved?

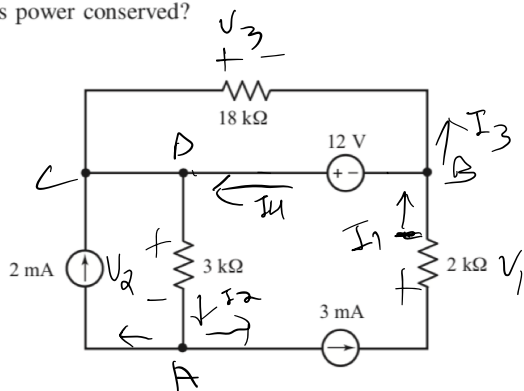


Figure P 3.2-28

(a) 12V Power supplied

$$P = IV \quad V = 12V \quad I_4 = 3 + \frac{2}{3} = \frac{11}{3} \text{ mA}$$

$$P = \left(\frac{11}{3} \times 10^{-3}\right) 12V = 44 \text{ mW}$$

2 mA

$$V_2 = 3 \times 5 = 15V$$

$$P = 2 \times 15 = 30 \text{ mW}$$

$$6V = 2 \times 3$$

$$V_1 = 15 - 6 = 9$$

$$P = 3 \times 9 = 27 \text{ mW}$$

(c) Power-supply = power-received

$$44 + 30 + 27 = 8 + 75 + 18 \quad \checkmark$$

Power is conserved.

(b) 2 kΩ Power Received

$$P = I_1^2 R = (3 \times 10^{-3})^2 \times 2,000$$

$$P = 18 \text{ mW}$$

3 kΩ

KCL @ Node A

$$I_2 = 3 \text{ mA} + 2 \text{ mA} = 5 \text{ mA}$$

$$P = (5 \times 10^{-3})^2 \times 3,000$$

$$= 75 \text{ mW}$$

18 kΩ

$$V_3 = 12V$$

$$I_3 = 12 / 18k = \frac{2}{3} \text{ mA}$$

$$P = \left(\frac{2}{3} \text{ mA}\right)^2 \times 18,000$$

$$= 8 \text{ mW}$$

P 3.3-10 Determine the value of the voltage measured by the meter in Figure P 3.3-10.

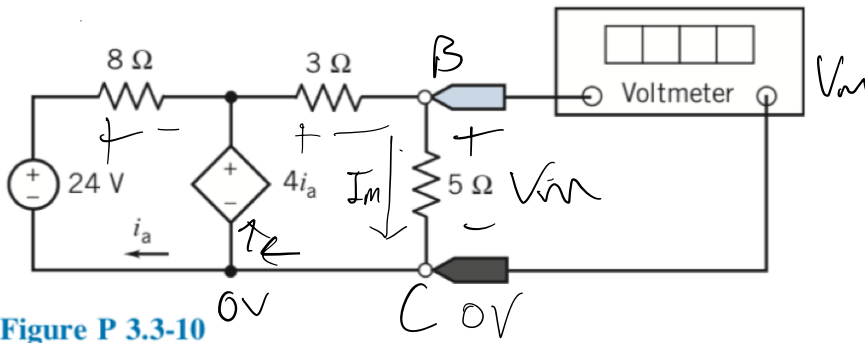


Figure P 3.3-10

Notes:

Current dependent Voltage Source

2 variables I_m , I_a

• Apply KCL + KVL

• circuit can be understood with mesh analysis.

KVL left mesh

$$-24 + 8i_a + 4i_a = 0 \quad i_a = 2A$$

KCL @ node A Right mesh.

$I_a + I_m \rightarrow$ Current flowing through 3Ω resistor.

KVL @ Right Loop

$$I_m(3\Omega) + I_m(5\Omega) - 8V = 0$$

$$I_m = 1A$$

$$V_m = 1 \times 5 = 5V$$

P 3.3-17 The input to the circuit shown in Figure P 3.3-17 is the voltage source voltage

$$v_s(t) = 12 \cos(377t) \text{ mV}$$

The output is the voltage $v_o(t)$. Determine $v_o(t)$.

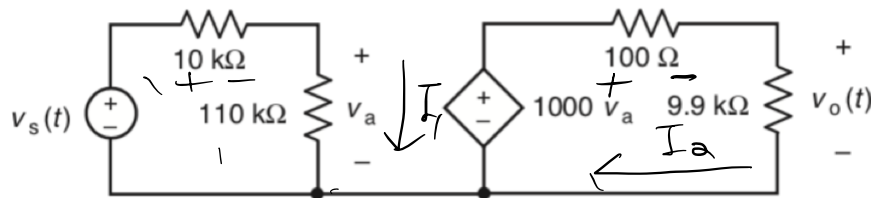


Figure P 3.3-17

$$V_{out} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_a = 12 \cos(377t) \times 10^{-3} \left(\frac{110k}{120k} \right)$$

$$V_a = 11 \cos(377t) \text{ mV}$$

$$V_o(t) = 1000 V_a \left(\frac{9900}{10,000} \right)$$

$$V_o(t) = 10.89 \cos(377t) \text{ V}$$

P 3.5-3 Determine the power received by each resistor in the circuit shown in Figure P 3.5-3.

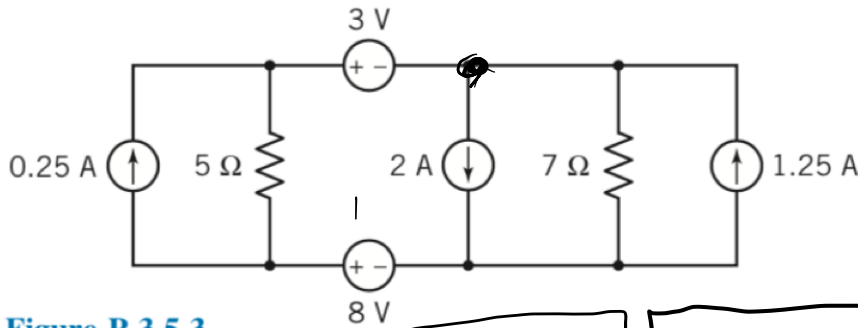
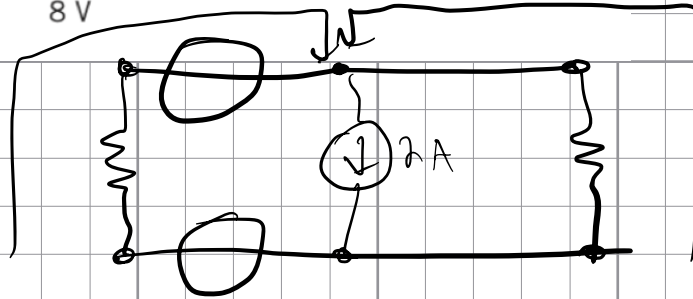


Figure P 3.5-3

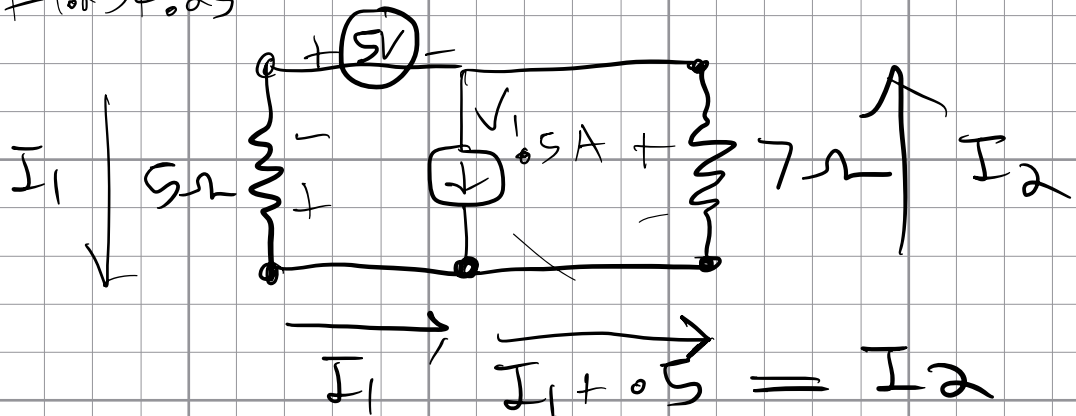


- This circuit is a current divider with 2 voltage sources in series.

- Combine sources and simplify circuit

$$I_T = -2A + 1.25 + 0.25$$

$$I_T = -0.5$$



KVL Left Mesh

$$+5 - 5I_1 = V_1$$

KVL Right Mesh

$$V_1 = 7I_2$$

$$P_{5\Omega} = 0.125^2 \times 5 = 7.8 \text{ mW}$$

$$P_{7\Omega} = 0.625^2 \times 7 = 2.73 \text{ W}$$

$$7I_2 + 5I_1 = 5$$

$$-I_2 + I_1 = -0.5$$

$$12I_1 = 1.5$$

$$I_1 = 0.125$$

$$I_2 = 0.625$$

P 6.3-1 + Determine the value of voltage measured by the voltmeter in Figure P 6.3-1.

Answer: -4 V

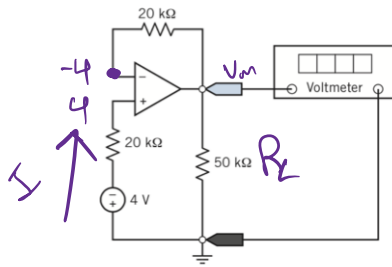


Figure P 6.3-1

The current out of negative side of amplifier is approx. zero.

Therefore, $20\text{ k}\Omega \times I \approx 0$

$$-4 - V_m = 0$$

$$V_m = -4\text{ V}$$

Home Work 5

P 6.3-3 + Find v_o and i_o for the circuit of Figure P 6.3-3.

Answer: $v_o = -30\text{ V}$ and $i_o = 3.5\text{ mA}$

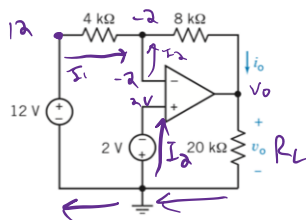


Figure P 6.3-3

KCL after $4\text{ k}\Omega$ resistor

$$I_1 = I_o - I_2$$

$$I_1 = 14/4\text{ k} = 3.5\text{ mA}$$

$$I_2 \approx 0\text{ mA}$$

$$I_o = 3.5\text{ mA}$$

$$-2 - v_o = 8 \times 3.5$$

$$-v_o = 28 + 2 = 30$$

$$v_o = -30\text{ V}$$

Problem #4

P 6.4-2 Find v_o and i_o for the circuit of Figure P 6.4-2.

Answer: $v_o = -4 \text{ V}$ and $i_o = 1.33 \text{ mA}$

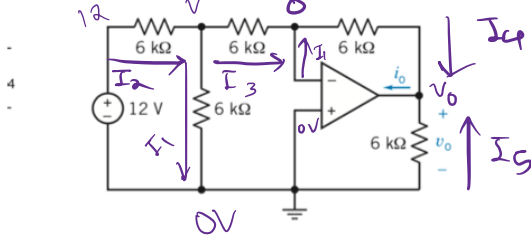


Figure P 6.4-2

Find

$$I_1 \approx 0$$

$$I_3 = I_4$$

$$I_4 = \frac{12 - v_o}{6k\Omega} - \frac{v_o}{6k\Omega} = I_3$$

$$I_3 = \frac{v_o}{6k\Omega} = \frac{12 - v_o}{6k\Omega} - \frac{v_o}{6k\Omega}$$

$$\frac{2v_o}{6k\Omega} = \frac{12 - v_o}{6k\Omega}$$

$$\frac{1}{3}v_o = 2 - \frac{1}{6}v_o \rightarrow v_o = 4 \text{ V}$$

$$I_4 = \frac{4}{6k\Omega} = \frac{2}{3} \text{ mA}, \text{ so}$$

$$0 - v_o = 6k\Omega \times \frac{2}{3} \text{ mA} \quad v_o = -4 \text{ V}$$

@ v_o KCL $I_o = \frac{2}{3} \text{ mA} + \frac{0 - v_o}{5k\Omega} = \frac{4}{3} \text{ mA}$

Home Work 5

Problem #5

Non-Ideal

P 6.7-1 Consider the inverting amplifier shown in Figure P 6.7-1. The operational amplifier is a typical OP-07E (Table 6.7-1). Use the offsets model of the operational amplifier to calculate the output offset voltage. (Recall that the input v_{in} is set to zero when calculating the output offset voltage.)

Answer: 0.45 mV

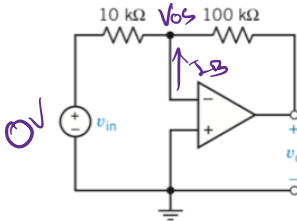


Figure P 6.7-1

• OP-07E offset - data

$$V_{os} = 30 \mu\text{V}$$

$$I_b = 1.2 \text{ nA}$$

KVL @ V_{os}

$$\frac{V_{os}}{10k\Omega} + \frac{V_{os} - v_o}{100k\Omega} + I_b = 0$$

$$\frac{v_o}{100k\Omega} = \frac{V_{os}}{10k\Omega} + \frac{V_{os}}{100k\Omega} + I_b$$

$$\begin{aligned} V_o &= \frac{30\mu\text{V}}{10k\Omega} + \frac{30\mu\text{V}}{100k\Omega} + 1.2\text{nA} \\ &= 3 \times 10^{-9} + 3 \times 10^{-3} + 1.2 \times 10^{-9} \\ &\quad \times 100k\Omega \\ &= 0.45 \text{ mV} \end{aligned}$$

P 3.6-11 Find i and $R_{eq\ a-b}$ if $v_{ab} = 40\text{ V}$ in the circuit of Figure P 3.6-11.

Answer: $R_{eq\ a-b} = 8\ \Omega$, $i = 5/6\text{ A}$

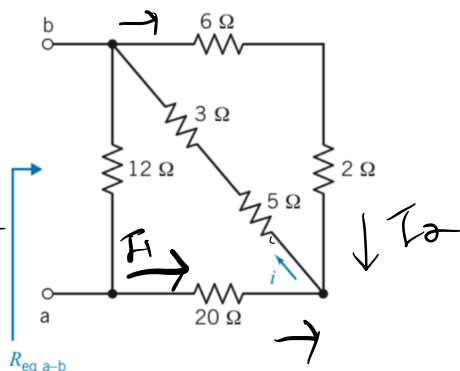
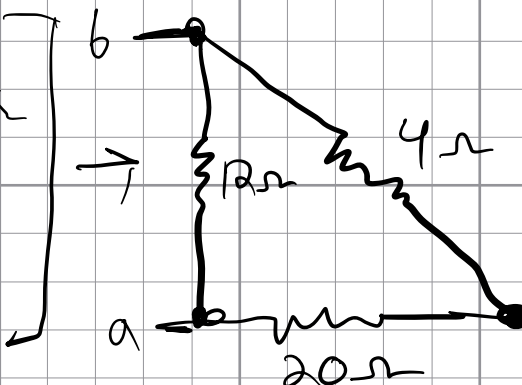


Figure P 3.6-11

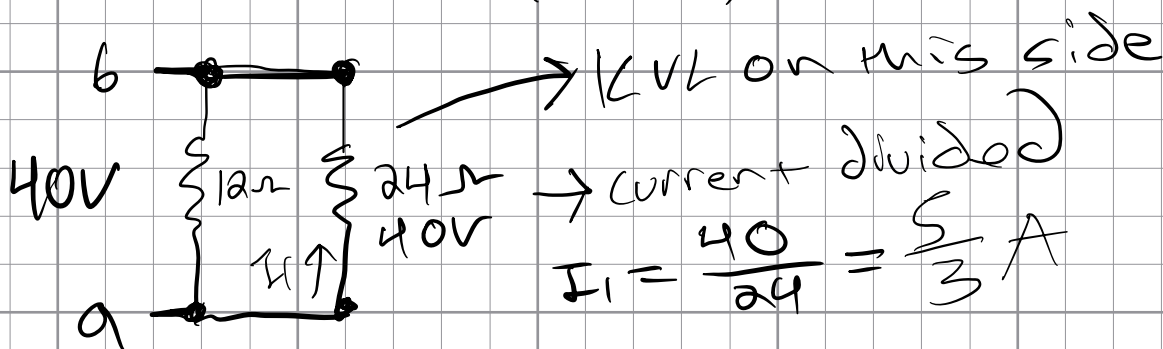
Ranvir Thind
05-09-26

1. $\frac{\text{Series}}{6+2=8\ \Omega}$ $\frac{\text{Series}}{3+5=8\ \Omega}$

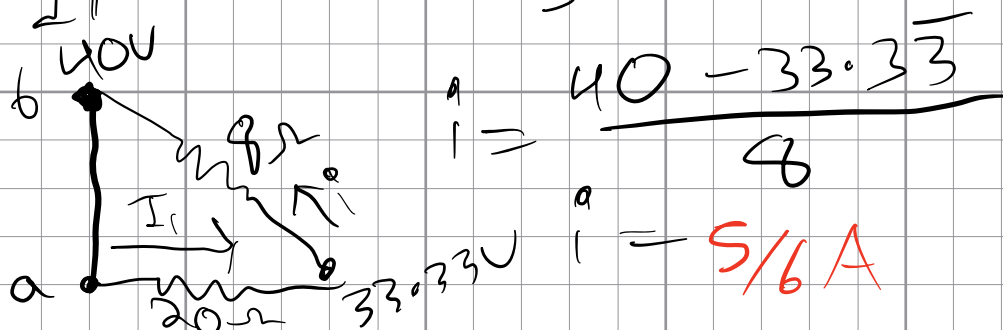
3. $\frac{\text{Parallel}}{\left(\frac{1}{8} + \frac{1}{8}\right)^{-1} = 4\ \Omega}$



4. $\frac{\text{Series}}{4+20=24\ \Omega}$ $\frac{\text{Parallel}}{R_{eq} = \left(\frac{1}{12} + \frac{1}{24}\right)^{-1} = 8\ \Omega}$



$I_1 = i - I_2 \Rightarrow \frac{5}{3} = i - I_2$



P 3.6-15 The circuit shown in Figure P 3.6-15 contains seven resistors, each having resistance R . The input to this circuit is the voltage source voltage v_s . The circuit has two outputs, v_a and v_b . Express each output as a function of the input.

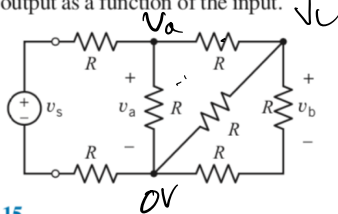
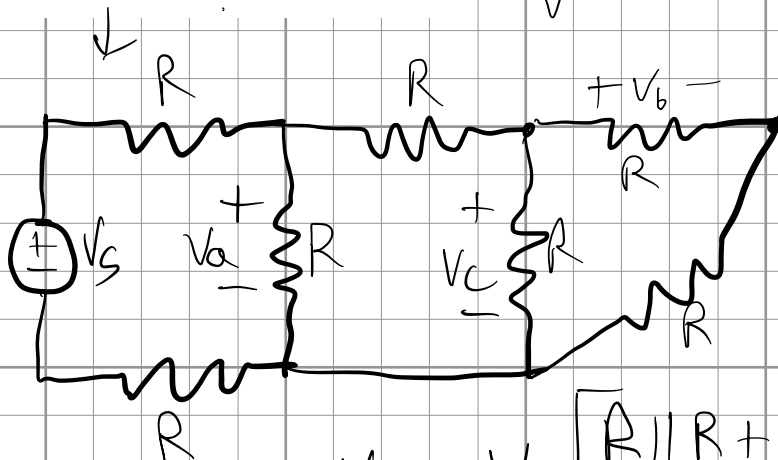


Figure P 3.6-15

$$\begin{matrix} v_a(V_s) \\ v_b(V_s) \end{matrix}$$

$$V_{out} = V_{in} \left(\frac{R_a}{R_a + R_1} \right)$$



$$V_a = V_s \frac{[R \parallel R + (R \parallel 2R)]}{2R + [R \parallel R + (R \parallel 2R)]}$$

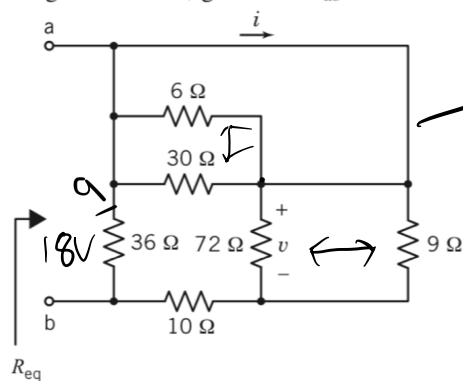
$$V_a = \frac{5}{21} V_s$$

$$V_c = V_a \frac{[R \parallel 2R]}{R + (R \parallel 2R)} = \frac{2}{5} V_a = \frac{2}{21} V_s$$

$$V_b = V_c \frac{R}{2R} = \frac{2}{21} V_s \left(\frac{1}{2} \right)$$

$$V_b = \frac{1}{21} V_s$$

P 3.6-20 Determine the values of i , v , and R_{eq} for the circuit shown in Figure P 3.6-20, given that $v_{ab} = 18$ V.



→ Parallel Short Circuit

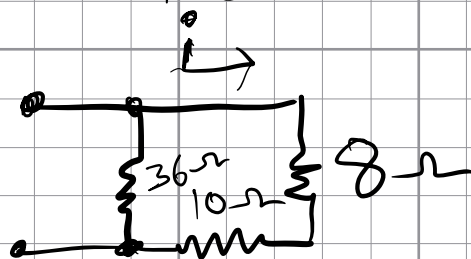


Figure P 3.6-20

$$V = 18 \frac{R_2}{R_1 + R_2} = \frac{\left(\frac{1}{72} + \frac{1}{9}\right)^{-1}}{10 + \left(\frac{1}{72} + \frac{1}{9}\right)^{-1}}$$

$$V = 18V \frac{8}{18} = 8V$$

$$R_{eq} = 10 + 8 \parallel 36 = 12\Omega$$

$$i = \frac{V}{R} = \frac{8}{8} = 1A$$

P 3.6-21 Determine the value of the resistance R in the circuit shown in Figure P 3.6-21, given that $R_{eq} = 9 \Omega$.

Answer: $R = 15 \Omega$

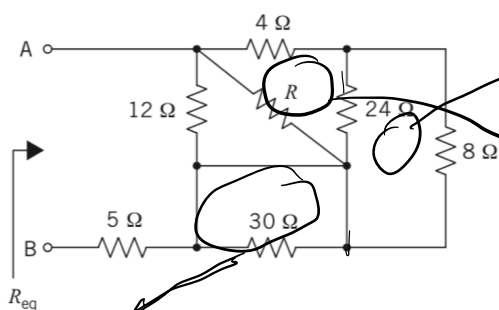


Figure P 3.6-21

Parallel

$$\left(\frac{1}{24} + \frac{1}{8} \right) = 6 \Omega$$

Series

$$6 + 4 = 10 \Omega$$

$$10 \Omega$$

Parallel

$$\left(\frac{1}{10} + \frac{1}{R} \right)^{-1}$$

$$12 \Omega$$

$$\frac{10R}{10+R}$$

Parallel

$$\left(\frac{1}{12} + \frac{10+R}{10R} \right)^{-1}$$

Short circuit || Resistor || Short circuit

Series

$$\frac{120R}{10R + 12R + 120} + 5 = R_{eq} = 9$$

$$120R = 88R + 480$$

$$R = 15 \Omega$$

P 3.6-21 Determine the value of the resistance R in the circuit shown in Figure P 3.6-21, given that $R_{eq} = 9 \, \Omega$.

Answer: $R = 15 \, \Omega$

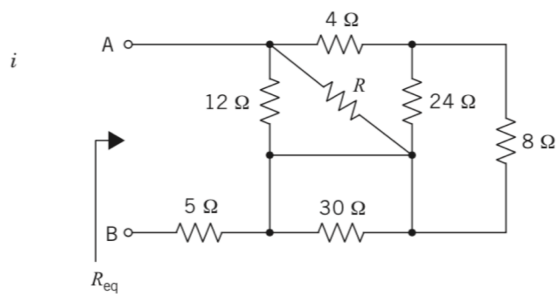


Figure P 3.6-21

Same as Last Problem

P 3.6-42 Determine the values of i_a , i_b , i_2 , and v_1 in the circuit shown in Figure P 3.6-42.

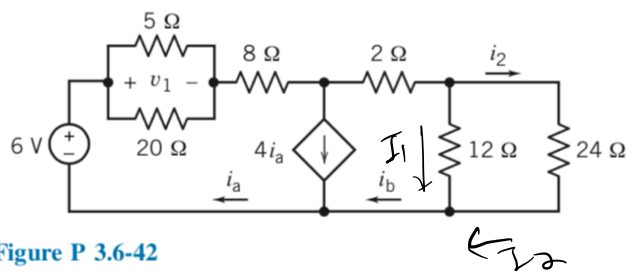
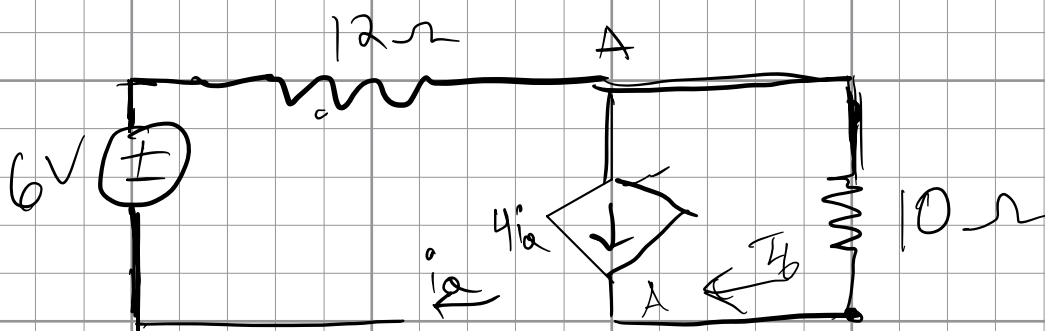


Figure P 3.6-42

1. Simplify/Rewrite circuit.

2. KVL + KCL



KCL @ A

$$I_b = -3i_a$$

KVL

$$-6 + 12i_a + 10i_b = 0$$

$$-18i_a = 6$$

$$-6 + 10 - \frac{8}{3} + v_1 = 0 \quad \begin{matrix} i_a = -\frac{1}{3} \text{ A} \\ i_b = 1 \text{ A} \end{matrix}$$

$$v_1 = -\frac{4}{3} \text{ V}$$

$$I_a = \frac{v}{R_{I_a}} = I_b \times \left(\frac{1}{12} + \frac{1}{24} \right)^{-1} / 24 \Omega$$

$$I_2 = \frac{1}{3} \text{ A}$$

P 3.6-47 Determine the values of the voltages v_1 and v_2 and of the current i_3 in the circuit shown in Figure P 3.6-47.

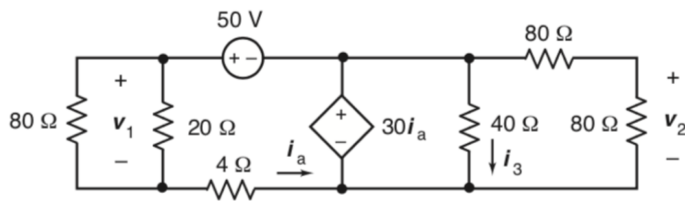
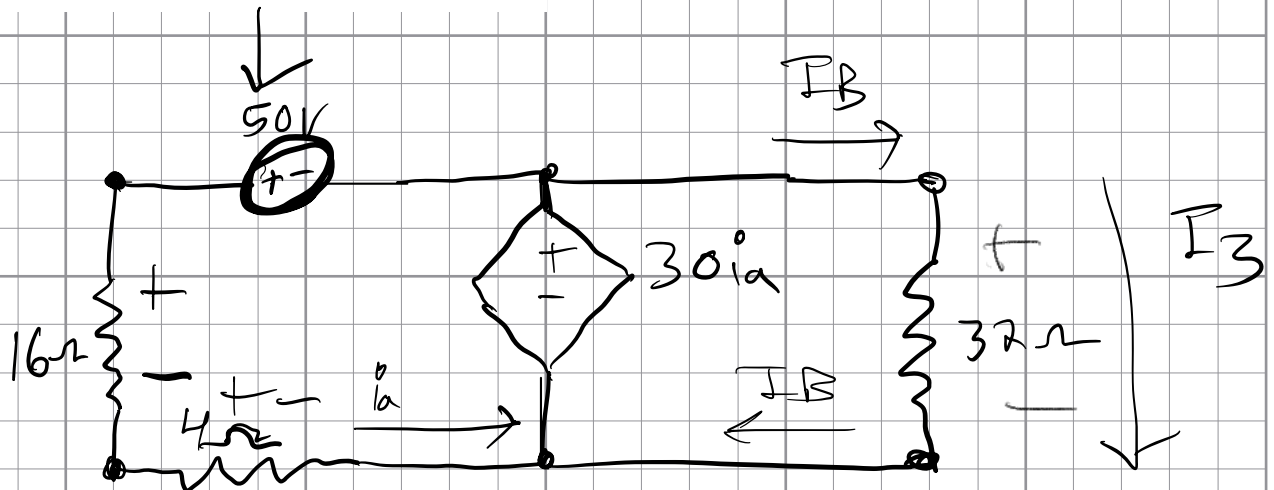


Figure P 3.6-47



Left Mesh KVL

$$-50 + 16i_a + 4i_a - 30i_a = 0$$

$$i_a = -5A \quad v_1 = 16 \times -5 = -80V$$

Right Mesh KVL

$$32I_B = 30i_a$$

$$I_B = -4.6875$$

$$v_2 = 30i_a \left(\frac{160}{40 + 160} \right) = -75V$$

$$I_3 = I_B \times \frac{160}{200} = -3.75V$$

P 3.6-23 Determine the values of r , the gain of the CCVS, and g , the gain of the VCCS, for the circuit shown in Figure P 3.6-23.

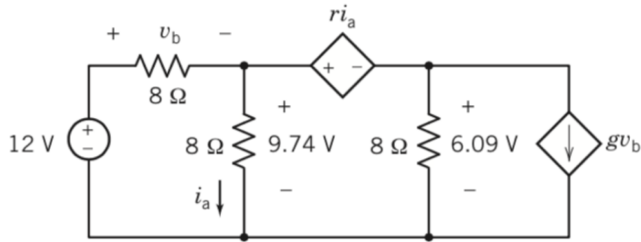


Figure P 3.6-23

$$i_a = 9.74 / 8 = 1.2175 \text{ A}$$

KVL $V_b = 12 - 9.74 \text{ V} = 2.26 \text{ V}$

$$r i_a = 9.74 - 6.09$$

$$r i_a = 3.65$$

$$r = 2.998 \text{ V/A}$$

KCL $g v_b + \frac{6.09}{8} - \frac{2.26}{8} + \frac{9.74}{8}$

$$g v_b = -1.696 \text{ A}$$

$$g = \frac{g v_b}{2.26 \text{ V}} = -0.75$$

DP 3-3 A phonograph pickup, stereo amplifier, and speaker are shown in Figure DP 3-3a and redrawn as a circuit model as shown in Figure DP 3-3b. Determine the resistance R so that the voltage v across the speaker is 16 V. Determine the power delivered to the speaker.

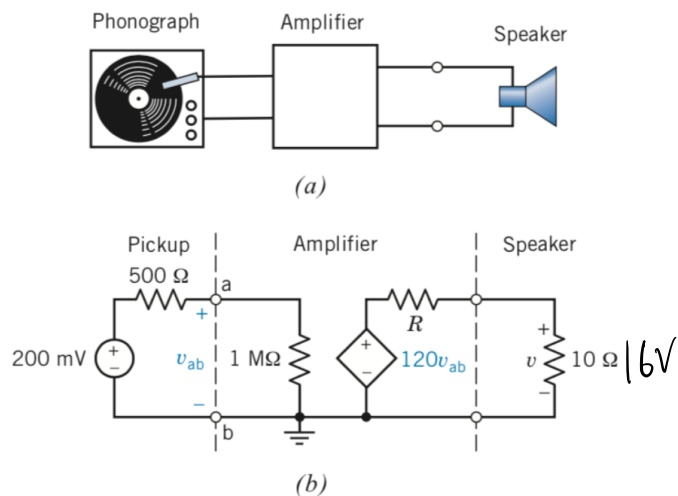


Figure DP 3-3 A phonograph stereo system.

$$V_{out} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_{ab} = 200 \text{ mV} \left(\frac{1 \text{ M}}{1 \text{ M} + 500} \right) = 199.9 \text{ mV}$$

$$V = 120 V_{ab} \left(\frac{10}{10 + R} \right)$$

$$16 = 120 \times 0.2 \left(\frac{10}{10 + R} \right)$$

$$R = 5 \Omega \quad I = \frac{16}{10} = 1.6$$

$$P_{5\Omega} = 1.6^2 \times 10 = 25.6 \text{ W}$$